Principal Component Analysis (application of the Spectral Thm)

Lecture 14

PCA is a widely used method of modern data analysis. It is one of the most valuable results from applied linear algebra.

~~[illegible]~~

<u>Problem</u>: Suppose we are trying to understand some phenomenon by measuring various quantities (velocities, humidity, income, etc) in our system. Suppose we measure m quantities and have n measurements. Let's assemble our data into a data matrix

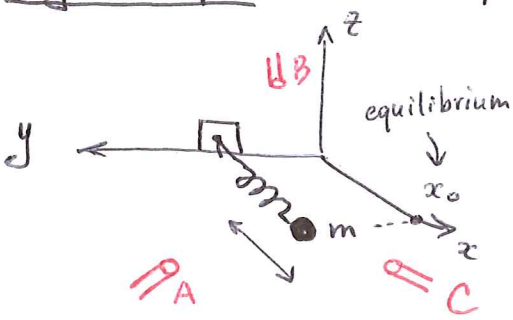
$$X = \begin{array}{c} \\ \\ \\ m \end{array} \underbrace{\begin{pmatrix} \cdots & x_j & \cdots \\ \cdots & \vdots & \cdots \\ \cdots & \vdots & \cdots \end{pmatrix}}_{n} \; x^i$$

$x^i = (x_{i1}, \dots, x_{in})$ the values of the ith quantity accross all measurements

$x_j = (x_{1j} \dots x_{mj})^T$ is the jth measurement.

x_{ij} is the values of the ith quantity in the jth measurement.

Let's consider the following <u>toy example</u>: simple harmonic oscillator (a ball of mass m attached to a massless, frictionless spring)



The dynamics is very simple:

$$\left[\begin{array}{ll} \bullet \text{ Newton's 2nd law} : & m\ddot{x} = F \\ \bullet \text{ Hooke's law} : & F = -k(x - x_0) \end{array}\right\} \Rightarrow x(t) = \underset{\text{equilibrium}}{x_0} + \underset{\text{amplitude}}{A} \cos\Big(\underbrace{\sqrt{\tfrac{k}{m}}}_{\frac{2\pi}{T}} t + \underset{\text{phase}}{\varphi}\Big)$$

But let's pretend that we don't know that this is an oscillator, and all we have is the data from three movie cameras, placed somewhere around our system (not necessarily along x, y, z axes). Each camera records two coordinates of the ball in the camera plane every Δt seconds

So our data matrix is

$$X = \begin{array}{c} \Delta t \; 2\Delta t \; \cdots\cdots \; n\Delta t \\ \begin{pmatrix} \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \\ \cdots & \cdots & \cdots & \cdots \end{pmatrix} \end{array} \begin{array}{l} x_A \\ y_A \\ x_B \\ y_B \\ x_C \\ y_C \end{array}$$

$m = 6$, $n = T/\Delta t$

total duration of the experiment

Ideally, we want $x_A(n\Delta t) = x(n\Delta t)$ and remaining rows are zero..

Q: How do we get from X to $x(t) = \dots$?

Or more generally:

we will be making this question more & more precise

$\boxed{Q1}$: Our measurements $x_1 \dots x_n \in \mathbb{R}^m$

Is there another (non-standard) basis of $\mathbb{R}^m$ that best ~~[illegible]~~ re-expresses the data X?

(to be discussed) (start from re-express)

Let $v_1 \dots v_m$ be another basis of $\mathbb{R}^m$, $v_j = \begin{bmatrix} v_{1j} \\ \vdots \\ v_{mj} \end{bmatrix}$ ← coordinates of v_j in the stand. basis.

Recall that coordinates $y_j = \begin{bmatrix} y_{1j} \\ \vdots \\ y_{mj} \end{bmatrix}$ of x_j in $\{v_j\}$ are related to coordinates $\begin{bmatrix} x_{1j} \\ \vdots \\ x_{mj} \end{bmatrix}$ of x_j in the standard basis as follows: $x_j = \underbrace{[v_1 \dots v_m]}_{V \in \mathbb{M}_{m \times m}} y_j$

Therefore, when we change basis $\{e_j\} \rightsquigarrow \{v_j\}$,

our data matrix X transforms to Y, where $X = VY$ or $\boxed{Y = V^{-1}X}$

data in $\{v_j\}$ — data in the stand. basis

So the question becomes:

$\boxed{Q2}$: Can we find V (basis) such that Y is "nice"?

Heads-up: $v_1 \dots v_m$ will become the principal components of X

To answer Q2, we need to formalize.

Q: What does "best" re-express the data mean? Or, what does "Y is nice" mean?

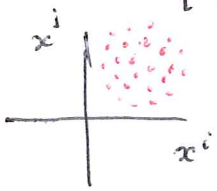
"Physics Perspective"

We often measure more than we need
↳ especially nowadays.

As a result, data often suffer from redundancy. This is particularly clear in our toy example: all cameras record the same dynamic information. Redundancy overcomplicates the picture and we would like to get rid of it.

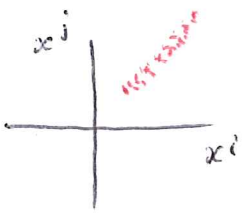
To get an intuitive fill for redundancy in the data, consider the following examples.

$$X = \begin{bmatrix} x^1 \\ \vdots \\ x^m \end{bmatrix} \begin{matrix} \leftarrow \\ \leftarrow \\ \leftarrow \\ \leftarrow \end{matrix}$$ measurement types (quantities) features



low redundancy: x^i and x^j are essentially uncorrelated

e.g. $x^i = x_A$
x^j = temperature

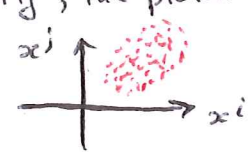


high redundancy: x^i and x^j are strongly correlated

e.g. $x^i = x_A$
$x^j = x_B$
A and B are close

These are two extreme cases.

In reality, the picture looks more like this:



x^i = high
x^j = weight

How to quantify the redundancy between two measurement types? x^i and x^j

We can use covariance.

Remark Correlation coeff may seem as a better option, but a) it is more complicated b) for our purposes, the two are equivalent.

Recall that if $x^i = (x_{i1}, \dots, x_{in})$ and $x^j = (x_{j1}, \dots, x_{jn})$, then covariance between x^i and x^j is (*)

$$\mathrm{Cov}(x^i, x^j) = \frac{1}{n} \sum_{k=1}^{n} x_{ik} x_{jk} - \left(\frac{1}{n} \sum_{k=1}^{n} x_{ik}\right)\left(\frac{1}{n} \sum_{k=1}^{n} x_{jk}\right)$$

If we assume that x^i and x^j have zero mean, $\frac{1}{n} \sum_{k=1}^{n} x_{ik} = \frac{1}{n} \sum_{k=1}^{n} x_{jk} = 0$,

then this simplifies to

$$\mathrm{Cov}(x^i, x^j) = \frac{1}{n} \sum x_{ik} x_{jk} = \frac{1}{n} x^i \cdot (x^j)^T$$

Recall that if ξ and η are two random variables, then

$$\mathrm{Cov}(\xi, \eta) = \mathbb{E}[\xi \cdot \eta] - \mathbb{E}[\xi]\,\mathbb{E}[\eta]$$

We can model x^i as a random variable and view

$x_{i1} \dots x_{in}$ as its realizations.

(*) is then simply the sample covariance between x^i and x^j

<u>Remark</u> We can always make x^i to have zero mean by transforming it $x^i \rightsquigarrow x^i - \frac{1}{n}\sum_{k=1}^{n} x_{ik}$. This does not affect covariance $\{x_1 \dots x_n\}$. Geometrically, this transformation simply shifts the point cloud.

In what follows, let's assume that rows of X have zero mean. (if not, we will pre-process the data by $x^i \rightsquigarrow x^i - \overline{x^i}$)

Covariance C_X does not change after this transformation. But transformation allows to write C_X like this

Def The covariance matrix of X is $\boxed{C_X = \frac{1}{n} X X^T}$

$$(C_X)_{ij} = \frac{1}{n} x^i (x^j)^T = \mathrm{Cov}(x^i; x^j).$$

So, the covariance matrix C_X quantifies the redundancy between all pairs of measurement types. To get rid of redundancy, we thus need to transform $X \rightsquigarrow Y$ such that C_Y is diagonal. (= "Y is nice" "Basis $v_1 \dots v_m$ is good")

$\boxed{Q3}$: Can we find principal components $v_1 \dots v_m$ such that C_Y is diagonal? where $Y = V^{-1} X$

Let's try to do this.

$$C_Y = \frac{1}{n} Y Y^T = \frac{1}{n} V^{-1} X X^T V^{-T} \quad V^{-1} C_X V^{-T}$$

This is true only if row sums of Y are zero. But this is indeed the case:

If $V^{-1} = (w_{ij})$, then

$$y_{ij} = \sum_k w_{ik} x_{kj} \quad \text{and}$$

$$\sum_j y_{ij} = \sum_j \left(\sum_k w_{ik} x_{kj}\right) = \sum_k w_{ik} \overbrace{\left(\sum_j x_{kj}\right)}^{0} = 0$$

Since $C_X = \frac{1}{n} X X^T$ is symmetric, according to the spectral theorem,

$$C_X = Q \Lambda Q^T, \text{ where}$$

- $\Lambda = \mathrm{diag}(\underbrace{\lambda_1 \dots \lambda_m}_{\text{eigenvalues of } C_X})$
- $Q = [\underbrace{q_1 \dots q_m}_{\text{corresp. eigenvectors of } C_X}]$ is orthogonal

Therefore, $C_Y = V^{-1} Q \Lambda Q^T V^{-T}$

To make C_Y diagonal, it is enough to set

$$\boxed{V = Q} \Rightarrow C_Y = \Lambda$$

So the principal components (directions) of X are the orthonormal eigenvectors of its covariance matrix $C_X = \frac{1}{n} X X^T$.

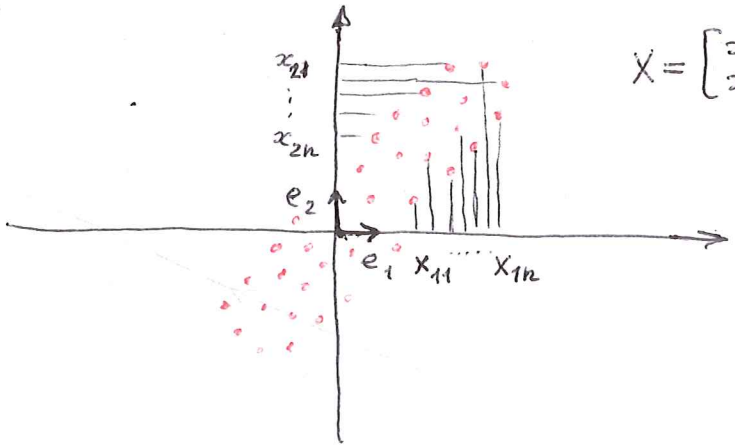
Statistical Interpretation of Principal Components (Using optimization interpretation of eigenvalues)

Principal component $v_i \equiv q_i$ is an eigenvector of C_X associated with eigenvalue λ_i

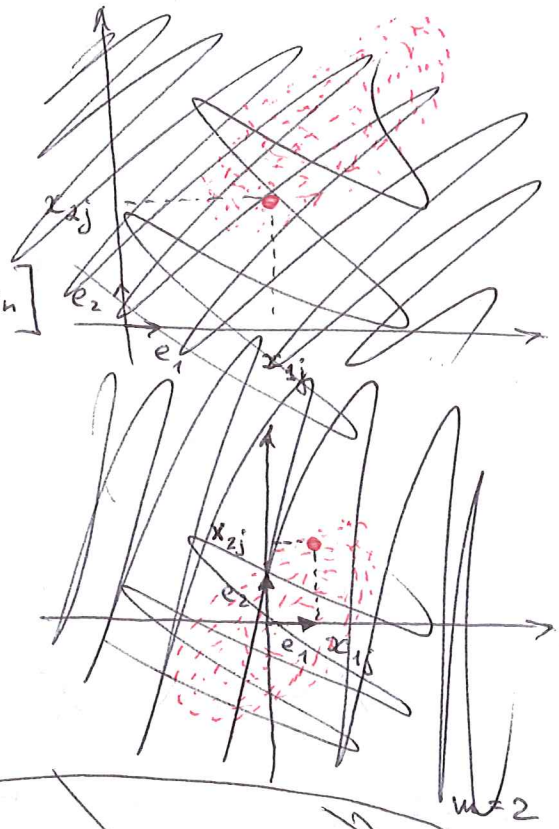
$\lambda_i = (C_Y)_{ii} = \frac{1}{n} y^i (y^i)^T = Var(y^i) = \sigma_i^2$, where

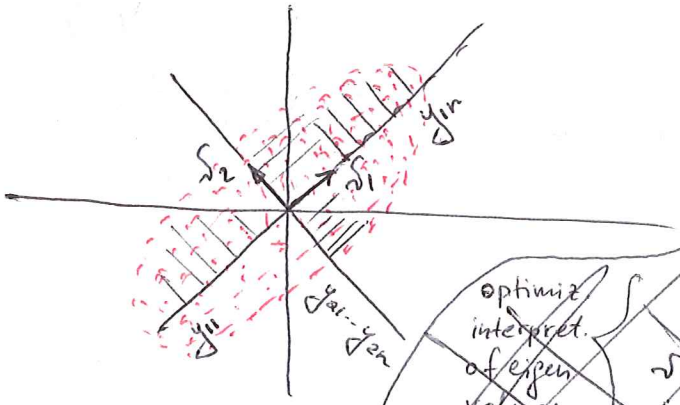
$y^i = (y_{i1} \cdots y_{in})$ are ~~transformed measurement types~~ transformed measurement types.

Here is how the picture looks for $m = 2$



$$X = \begin{bmatrix} x^1 \\ x^2 \end{bmatrix} = [x_1 \dots x_n]$$





Assume that $\lambda_1 \geq \dots \geq \lambda_n$

or $\sigma_1^2 \geq \dots \geq \sigma_n^2$

v_1 is associated with the largest variance σ_1^2

v_2 is associate with the second largest variance σ_2^2

$\vdots$

$v_i \in span(v_1 \dots v_{i-1})^{\perp}$ is assoc. with variance σ_i^2

etc.

optimiz. interpret. of eigen values. (next time)

To sum up

Given X, we find its principal components:

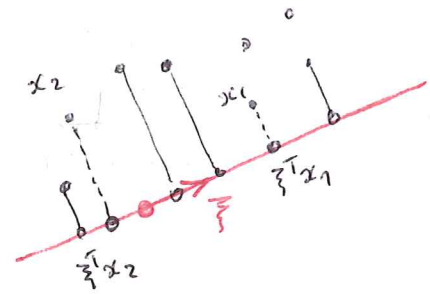
1. Center X: $x^i \rightsquigarrow x^i - \overline{x^i}$
2. ~~Find Diagonalize~~ Find an orthonormal eigen basis of the covariance matrix $C_X = \frac{1}{n} X X^T$.

Assume that $\lambda_1 \dots \lambda_m$ are ordered so that $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m$,
or equivalently $\sigma_1^2 \geq \sigma_2^2 \geq \dots \geq \sigma_m^2$

Recall that $\lambda_1 = \max\limits_{\xi \in S^m} \xi^T C_X \xi$ and the maximum is achieved on $\xi \in S^n \cap V_{\lambda_1}$, in particular on v_1

$$\sigma_1^2 = \lambda_1 = \max_{\|\xi\|=1} \xi^T \tfrac{1}{n} X X^T \xi = \max_{\|\xi\|=1} \tfrac{1}{n} (\xi^T X)(\xi^T X)^T = \max_{\|\xi\|=1} Var(\xi^T X)$$

$$\xi^T X = \xi^T [x_1 \dots x_n] = (\underbrace{\xi^T x_1, \dots, \xi^T x_n}_{\text{projections of } x_1 \dots x_n \text{ on } \xi})$$



This maximum is achieved when $\xi = v_1$ and

$$\xi^T X = v_1^T X = y^1$$

$$Y = V^{-1} X = V^T X$$

$$\begin{bmatrix} y^1 \\ \vdots \\ y^m \end{bmatrix} = \begin{bmatrix} v_1^T \\ \vdots \\ v_m^T \end{bmatrix} [x_1 \dots x_n]$$

Thus: v_1 is the direction such that the variance of projections of $x_1 \dots x_n$ on v_1 is maximized.

$v_2 \in span(v_1)^\perp$ is the —"—